Advanced Functions

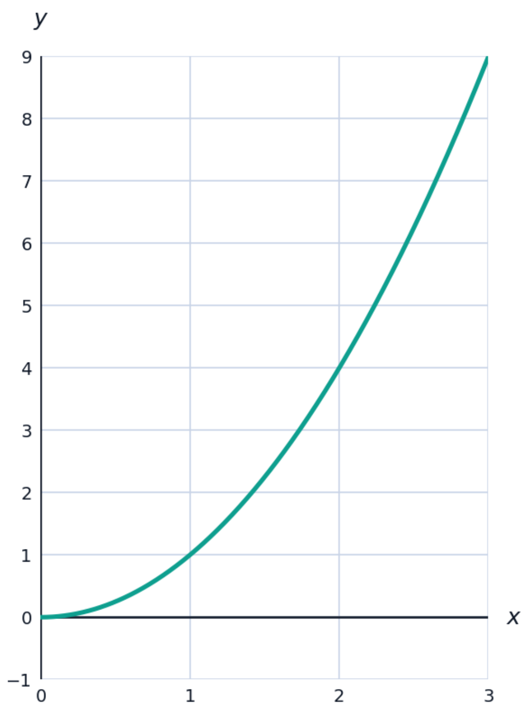
Composite functions with restrictions

1 If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

2 If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

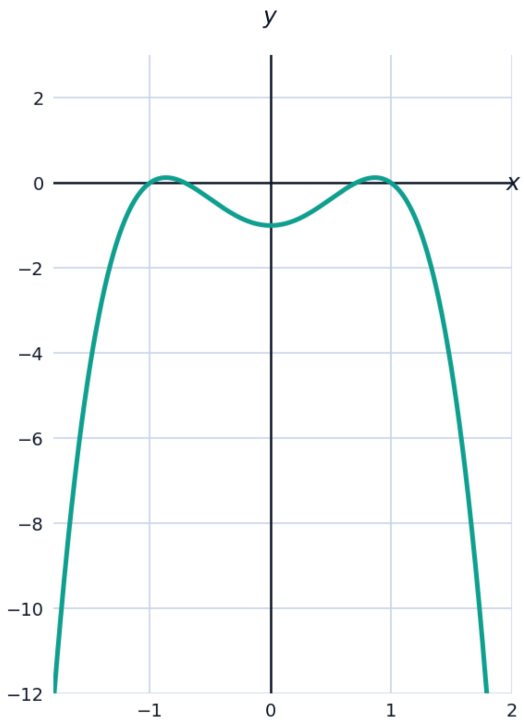
Inverse functions with domain restrictions

3 Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.

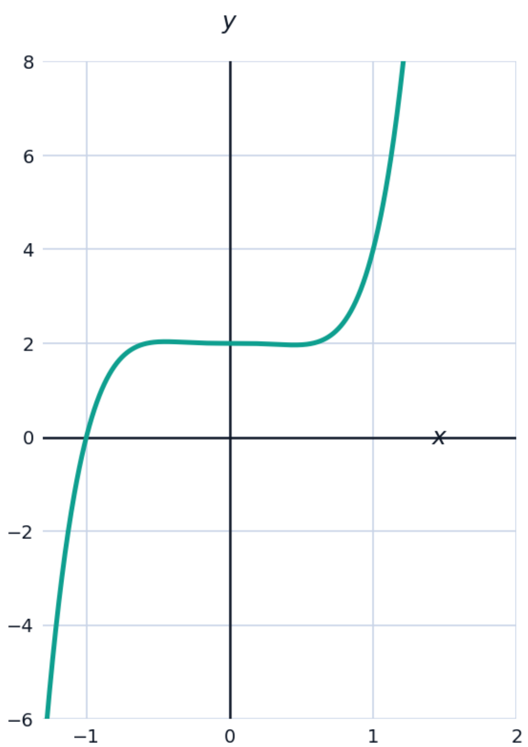


End behavior of polynomial functions

4 Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.

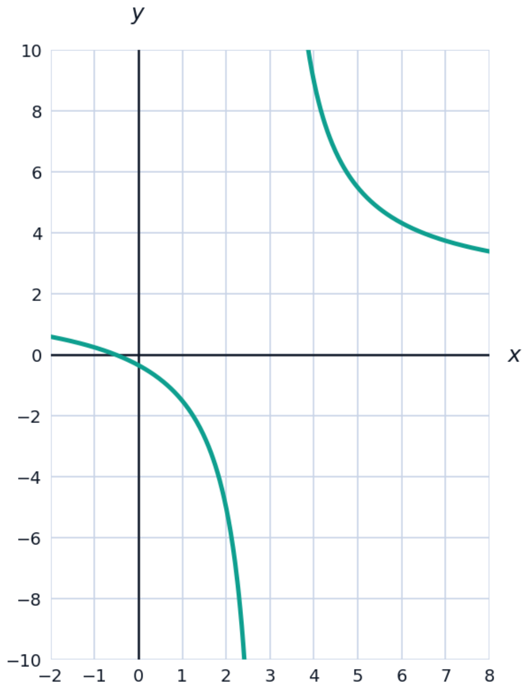


5 Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.



Rational functions: asymptotes and holes

6 Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.



7 Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2-4}{x-2}$.

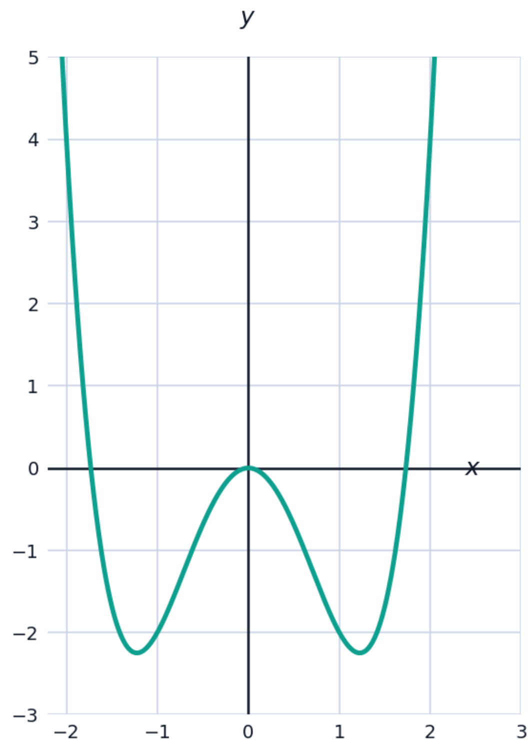
Logarithmic functions and their properties

8 Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

9 Solve $\ln(x) + \ln(x-3) = \ln(10)$.

Even, odd, and periodic function classification

10 Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.



Piecewise-defined functions and continuity

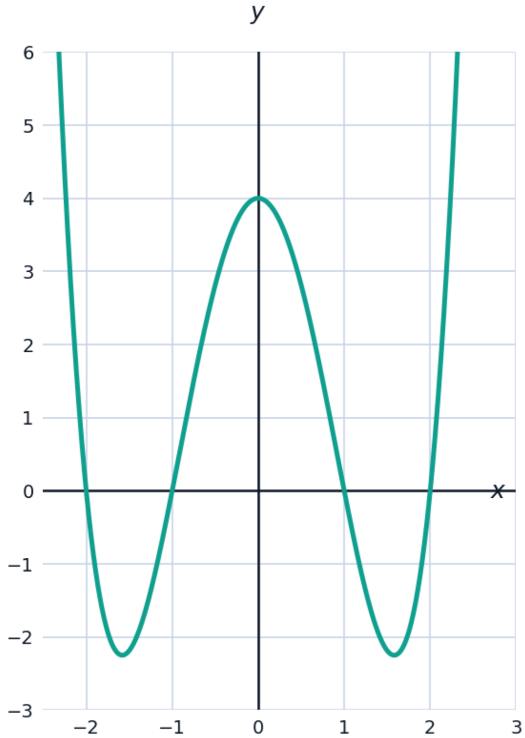
11 A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.

A multi-part function transformation and composition problem

12 This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.

Finding zeros of higher-degree functions graphically and algebraically

13 A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

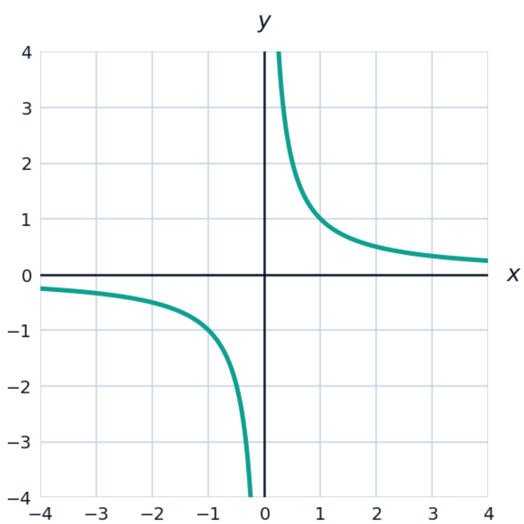


Function operations: sum, difference, product, quotient

14 Given $f(x) = x^2 + 1$ and $g(x) = x - 3$, find $(f+g)(x)$, $(f-g)(x)$, and $(fg)(x)$.

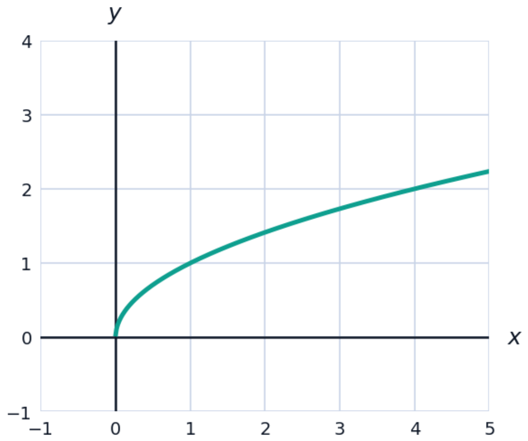
Symmetry and transformations of rational functions

15 Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.



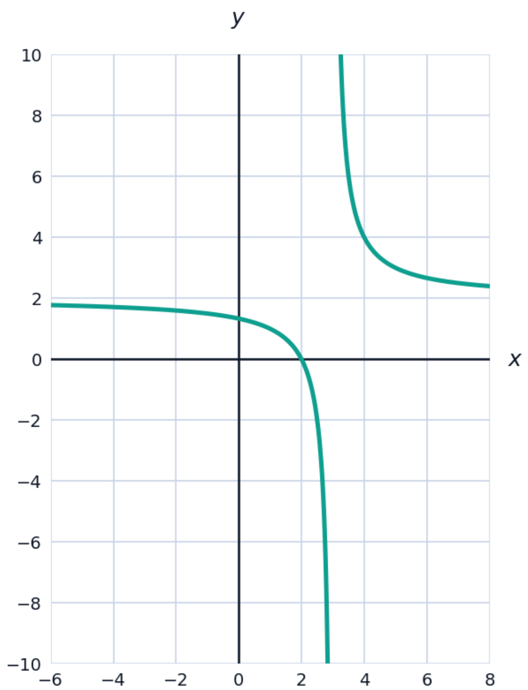
Finding a function's formula from a description of transformations

16 The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.



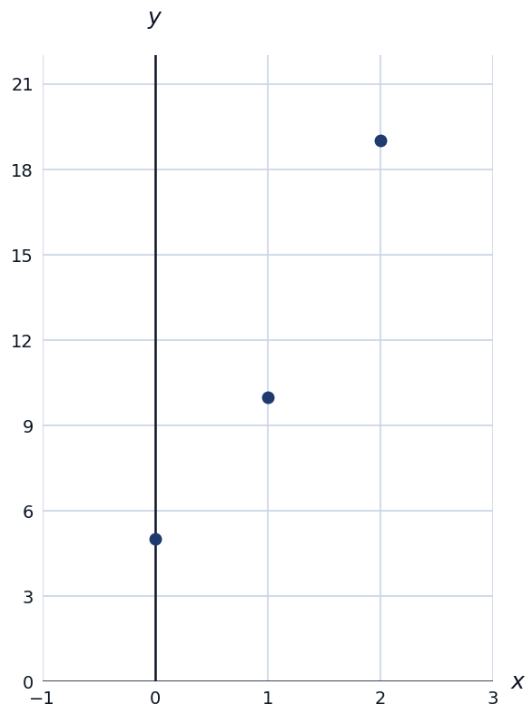
A multi-part rational function analysis problem

17 This question has three parts. A function is given by $f(x) = \dfrac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.



Finding a quadratic function from a table of values

18 A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a, b, and c.



Determining if a function has an inverse using the horizontal line test

19 Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.